# AI/ML intern DAY 3

## Task 3 AI RESEARCH

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## 1. Virtual Try-On AI Systems

### Introduction

Virtual try-on systems are AI-powered applications that allow users to digitally try clothes, shoes, glasses, or accessories before purchasing them online.

These systems combine:

- Artificial Intelligence
- Computer Vision
- Augmented Reality
- Pose Detection
- Deep Learning

to create an interactive shopping experience.

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## How Virtual Try-On Systems Work

### Step 1 — Human Body Detection

The AI system detects the user’s body using cameras or uploaded images.

### Step 2 — Pose Detection

Body landmarks such as shoulders, hips, and arms are identified.

### Step 3 — Body Segmentation

The system separates the human body from the background.

### Step 4 — Clothing Overlay

Virtual clothing is digitally placed on the body.

### Step 5 — Real-Time Visualization

The user can see how the clothing appears from different angles.

**Technologies Used**

Virtual try-on systems commonly use:

- OpenCV
- MediaPipe
- TensorFlow
- Augmented Reality
- Deep Learning
- Computer Vision

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**Applications**

Applications include:

- fashion e-commerce
- online clothing stores
- eyewear try-on systems
- makeup applications
- footwear shopping

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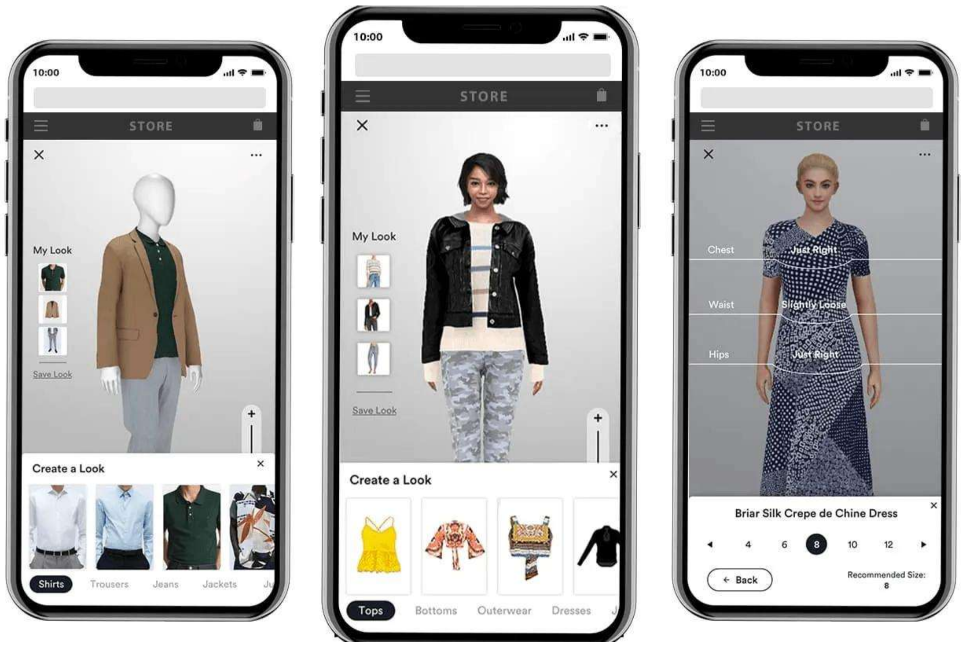
**Advantages**

Benefits include:

- improved customer experience
- reduced product returns
- accurate size visualization
- increased customer confidence
- personalized shopping




## 2. Fashion Recommendation Models

### Introduction

Fashion recommendation models are AI systems that suggest clothing and fashion products based on customer preferences, behaviour, and shopping history.

These systems help customers discover products more efficiently and improve online shopping personalization.

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### How Fashion Recommendation Models Work

AI recommendation systems analyse:

- purchase history
- browsing behaviour
- fashion interests
- trending products
- customer preferences

Machine learning algorithms identify patterns and recommend suitable fashion products.

### Types of Recommendations

#### Personalized Recommendations

Suggestions based on individual customer behaviour.

#### Trending Recommendations

Products currently popular among customers.

#### Seasonal Recommendations

Suggestions based on weather and seasons.

#### Outfit Matching

AI suggests matching clothes and accessories.

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### Technologies Used

Fashion recommendation systems use:

- Machine Learning
- Deep Learning
- Recommendation Algorithms
- Customer Data Analysis
- Artificial Intelligence

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### Applications

Applications include:

- online fashion stores
- fashion apps
- e-commerce websites
- AI fashion assistants
- personalized shopping platforms

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**Advantages**

Benefits include:

- faster product discovery
- personalized shopping
- increased customer satisfaction
- improved sales
- better user engagement

t-SNE representation of some dresses and shoes projected into style space. The blue box highlights an area containing occasion wear while the red box contains casual day wear.